

ConnectEU — System Design & Product Requirements (MVP)

Overview

ConnectEU is a social media platform specifically designed for users within the European Union.

The platform focuses on:

- Secure communication
- Community engagement
- GDPR compliance
- Transparent data handling

Users can:

- Create profiles
- Post text, images, and videos
- Follow users
- Send direct messages
- Join community groups

Goal:

Deliver a scalable, secure, and regulation-compliant social platform for EU users.

PART 1: FUNCTIONAL REQUIREMENTS

1. User Authentication & Profile Management

Description:

Users can sign up, log in, edit profiles, and manage account settings.

Impact Area:

Both Frontend and Backend

2. Content Posting

Description:

Users can create text, image, and video posts.

Impact Area:

Both Frontend and Backend

3. Feed Generation

Description:

Users can view posts from followed users and groups.

Impact Area:

Backend

4. Follow/Unfollow Users

Description:

Users can follow creators and other users.

Impact Area:

Both Frontend and Backend

5. Direct Messaging

Description:

Users can send and receive private messages.

Impact Area:

Both Frontend and Backend

6. Community Groups

Description:

Users can create, join, and interact in groups.

Impact Area:

Both Frontend and Backend

7. Notifications System

Description:

Users receive notifications for likes, comments, follows, and messages.

Impact Area:

Backend

PART 2: NON-FUNCTIONAL REQUIREMENTS

1. GDPR Compliance

- User consent management
 - Right to delete data
 - Data portability
 - Privacy controls
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2. Data Residency

- All EU user data should remain within EU data centers
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3. Scalability

- Support millions of users and concurrent requests
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4. Security

- End-to-end encryption
 - Secure authentication
 - DDoS protection
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5. High Availability

- Maintain 99.9% uptime
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6. Performance

- Feed load time below 2 seconds
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7. Content Moderation

- AI-assisted moderation and abuse detection

PART 3: DATA MODEL DESIGN

USER PROFILE MODULE

Entity: User

Attributes:

- user_id (UUID)
- username
- email
- password_hash
- bio
- profile_picture_url
- country
- created_at
- privacy_settings

Entity: Follow

Attributes:

- follow_id
- follower_user_id
- following_user_id
- created_at

Relationships:

- One User can follow many Users
- One User can have many followers

CONTENT POSTING MODULE

Entity: Post

Attributes:

- post_id
- user_id

- content_text
 - media_url
 - media_type (image/video/text)
 - visibility
 - created_at
 - updated_at
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Entity: Comment

Attributes:

- comment_id
 - post_id
 - user_id
 - comment_text
 - created_at
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Entity: Like

Attributes:

- like_id
 - user_id
 - post_id
 - created_at
-

Entity: Group

Attributes:

- group_id
 - group_name
 - description
 - owner_user_id
 - created_at
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Relationships:

- One User can create many Posts
- One Post can have many Comments

- One Post can have many Likes
- One User can join many Groups
- One Group can contain many Posts

PART 4: MVP API DESIGN

1. User Signup API

Method: POST

Endpoint: /api/v1/auth/signup

Input:

- Name
- Email
- Password

Output:

- JWT Token
- User ID

Supports:

Authentication

2. User Login API

Method: POST

Endpoint: /api/v1/auth/login

Input:

- Email
- Password

Output:

- JWT Token

Supports:

Authentication

3. Create Post API

Method: POST

Endpoint: /api/v1/posts

Input:

- Text
- Media

Output:

- Post Object

Supports:

Content Posting

4. Get Feed API

Method: GET

Endpoint: /api/v1/feed

Input:

- User ID

Output:

- Feed Posts List

Supports:

Feed Generation

5. Follow User API

Method: POST

Endpoint: /api/v1/follow

Input:

- follower_id
- following_id

Output:

- Success Status

Supports:

Follow System

6. Send Message API

Method: POST

Endpoint: /api/v1/messages

Input:

- sender_id
- receiver_id
- message

Output:

- Message Object

Supports:

Direct Messaging

7. Create Group API

Method: POST

Endpoint: /api/v1/groups

Input:

- Group Name
- Description

Output:

- Group Object

Supports:

Community Groups

8. Join Group API

Method: POST

Endpoint: /api/v1/groups/{id}/join

Input:

- User ID

Output:

- Membership Status

Supports:
Community Groups

9. Add Comment API
Method: POST
Endpoint: /api/v1/posts/{id}/comments

Input:
- Comment Text

Output:
- Comment Object

Supports:
Engagement

10. Like Post API
Method: POST
Endpoint: /api/v1/posts/{id}/like

Input:
- User ID

Output:
- Like Status

Supports:
Engagement

HIGH-LEVEL ARCHITECTURE OVERVIEW

Frontend:
- React / Flutter Web Application
- Responsive UI
- API Gateway Integration

Backend:

- Authentication Service
 - Feed Service
 - Messaging Service
 - Media Upload Service
 - Notification Service
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Database:

- PostgreSQL for relational data
 - Redis for caching
 - Object Storage for media files
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Infrastructure:

- EU-hosted cloud infrastructure
 - CDN for media delivery
 - Kubernetes for scalability
 - Load balancers for high availability
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SECURITY & COMPLIANCE CONSIDERATIONS

GDPR Compliance Features:

- User consent management
 - Right to delete account/data
 - Download personal data
 - Cookie consent tracking
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Security Measures:

- OAuth2 / JWT authentication
 - Encrypted passwords
 - HTTPS everywhere
 - Rate limiting
 - Audit logging
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SCALABILITY STRATEGY

To support future growth:

- Use microservices architecture
- Horizontal scaling for APIs
- Queue-based notification delivery
- Feed caching using Redis
- CDN-based media optimization

FINAL RECOMMENDATION

For the MVP launch, ConnectEU should prioritize:

1. Secure authentication
2. Fast content sharing
3. Reliable feed generation
4. Privacy-first architecture
5. GDPR compliance from Day 1

The platform's competitive advantage in the EU market will come from:

- Trust
- Data privacy
- Transparency
- Community-driven engagement
- Reliable user experience